exploratory_analysis

March 17, 2026

1 Milestone 1 Data Analysis

1.1 Comparison of Optimization Strategies for Logistic Regression

This notebook supports the Milestone report for the capstone project. It performs:

- basic exploratory data analysis on the Iris dataset
- data preprocessing and standardization
- training with BGD, SGD, and MBGD
- comparison of convergence, accuracy, and computational cost
- a small learning-rate sensitivity experiment

1.2 1. Imports

[1]:
```
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import load_iris

# Make sure the project root is importable when this notebook is run from /
↪notebooks
PROJECT_ROOT = Path.cwd().resolve().parent
if str(PROJECT_ROOT) not in sys.path:
    sys.path.append(str(PROJECT_ROOT))

from data_loader import load_data
from train import train_model
from evaluation import compare_runs
from plots import plot_loss_curves, plot_accuracy_curves

plt.rcParams['figure.figsize'] = (8, 5)
```

1.3 2. Load the Iris dataset for EDA

[2]:
```
iris = load_iris(as_frame=True)
df = iris.frame.copy()
df['target_name'] = df['target'].map(dict(enumerate(iris.target_names)))

df.head()
```

[2]:
```
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  \
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2

   target target_name
0       0      setosa
1       0      setosa
2       0      setosa
3       0      setosa
4       0      setosa
```

1.4 3. Basic dataset inspection

[3]:
```
print('Shape:', df.shape)
print('\nColumn names:')
print(df.columns.tolist())
print('\nMissing values per column:')
print(df.isnull().sum())
print('\nClass distribution:')
print(df['target_name'].value_counts())
```

```
Shape: (150, 6)

Column names:
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width
(cm)', 'target', 'target_name']

Missing values per column:
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
target_name          0
dtype: int64

Class distribution:
```

```
target_name
setosa        50
versicolor    50
virginica     50
Name: count, dtype: int64
```

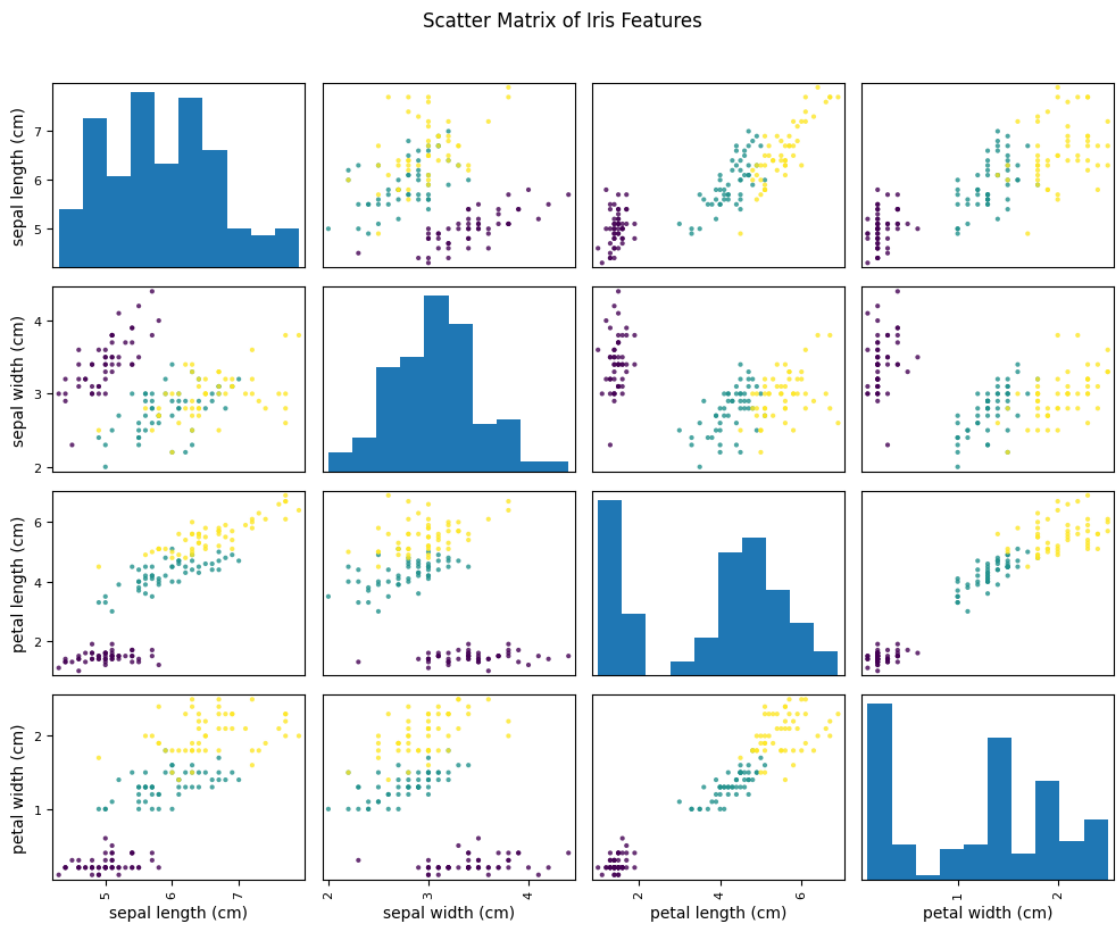
1.5 4. Summary statistics

[4]:
```
df.describe()
```

[4]:
```
       sepal length (cm)  sepal width (cm)  petal length (cm)  \
count         150.000000        150.000000         150.000000
mean            5.843333          3.057333           3.758000
std             0.828066          0.435866           1.765298
min             4.300000          2.000000           1.000000
25%             5.100000          2.800000           1.600000
50%             5.800000          3.000000           4.350000
75%             6.400000          3.300000           5.100000
max             7.900000          4.400000           6.900000

       petal width (cm)      target
count        150.000000  150.000000
mean           1.199333    1.000000
std            0.762238    0.819232
min            0.100000    0.000000
25%            0.300000    0.000000
50%            1.300000    1.000000
75%            1.800000    2.000000
max            2.500000    2.000000
```

1.6 5. Pairwise visualization

[5]:
```
pd.plotting.scatter_matrix(
    df.iloc[:, :4],
    c=df['target'],
    figsize=(10, 8),
    diagonal='hist',
    alpha=0.8,
)
plt.suptitle('Scatter Matrix of Iris Features', y=1.02)
plt.tight_layout()
plt.show()
```

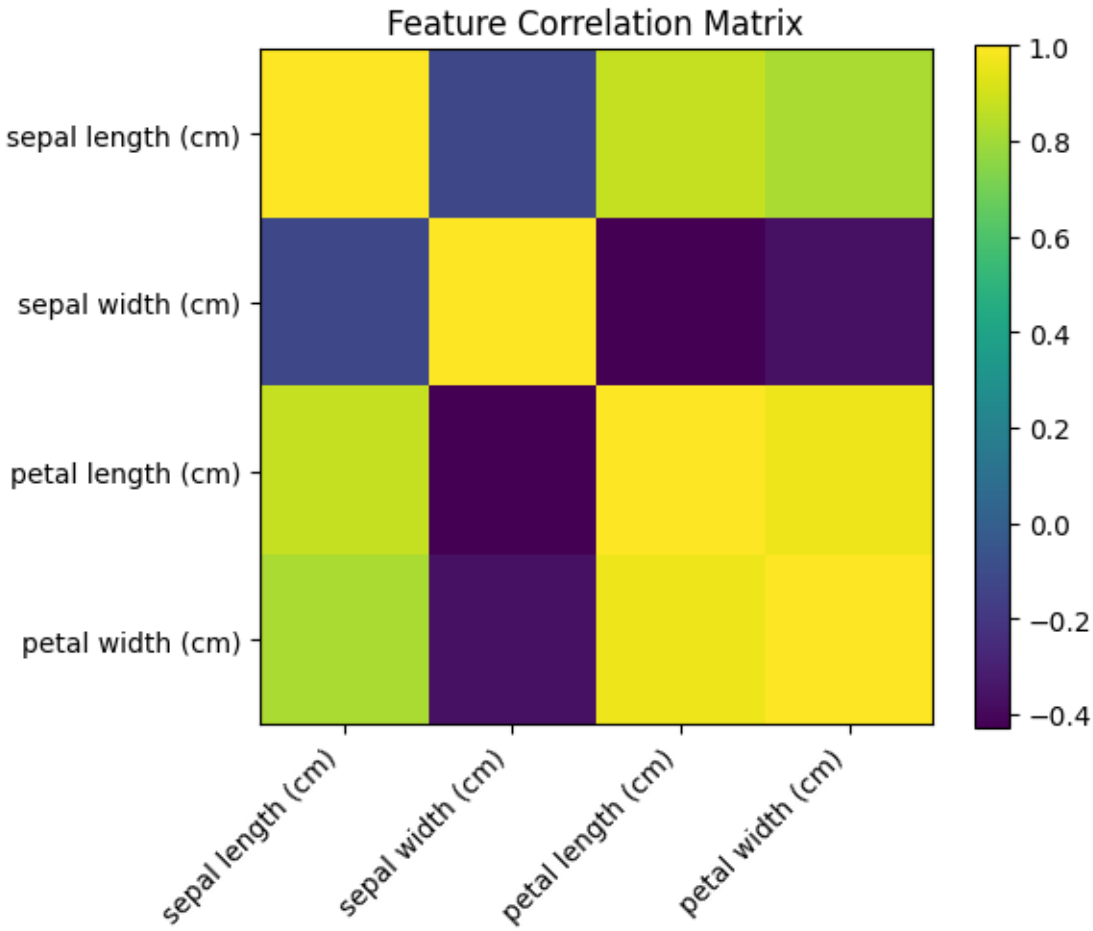


1.7 6. Correlation analysis

[6]:
```
corr = df.iloc[:, :4].corr()
corr
```

[6]:
```
                   sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)           1.000000         -0.117570           0.871754
sepal width (cm)           -0.117570          1.000000          -0.428440
petal length (cm)           0.871754         -0.428440           1.000000
petal width (cm)            0.817941         -0.366126           0.962865

                   petal width (cm)
sepal length (cm)          0.817941
sepal width (cm)          -0.366126
petal length (cm)          0.962865
petal width (cm)           1.000000
```

[7]:
```
plt.figure(figsize=(6, 5))
plt.imshow(corr, interpolation='nearest')
plt.xticks(range(len(corr.columns)), corr.columns, rotation=45, ha='right')
plt.yticks(range(len(corr.index)), corr.index)
plt.colorbar()
plt.title('Feature Correlation Matrix')
plt.tight_layout()
plt.show()
```



1.8 7. Load standardized train/test data

[8]:
```
X_train, X_test, y_train, y_test = load_data(test_size=0.2, random_state=42)

print('Training shape:', X_train.shape)
print('Test shape:', X_test.shape)
print('Training class counts:', np.bincount(y_train))
print('Test class counts:', np.bincount(y_test))
```

```
Training shape: (120, 4)
Test shape: (30, 4)
Training class counts: [40 40 40]
Test class counts: [10 10 10]
```

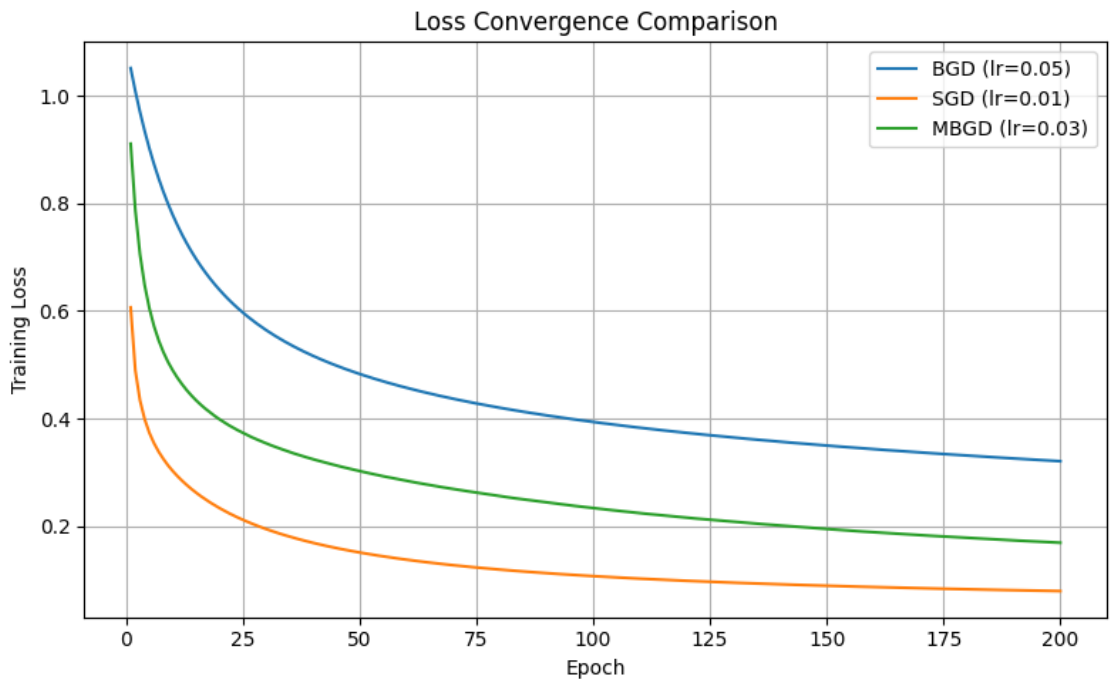
1.9 8. Main experiment: BGD vs SGD vs MBGD

[9]:
```
settings = [
    {'optimizer': 'bgd', 'learning_rate': 0.05, 'epochs': 200, 'batch_size':␣
 ↪16},
    {'optimizer': 'sgd', 'learning_rate': 0.01, 'epochs': 200, 'batch_size': 1},
    {'optimizer': 'mbgd', 'learning_rate': 0.03, 'epochs': 200, 'batch_size':␣
 ↪16},
]

histories = []

for cfg in settings:
    _, history = train_model(
        X_train,
        y_train,
        X_test,
        y_test,
        optimizer=cfg['optimizer'],
        learning_rate=cfg['learning_rate'],
        epochs=cfg['epochs'],
        batch_size=cfg['batch_size'],
    )
    histories.append(history)

summary_df = compare_runs(histories)
summary_df
```

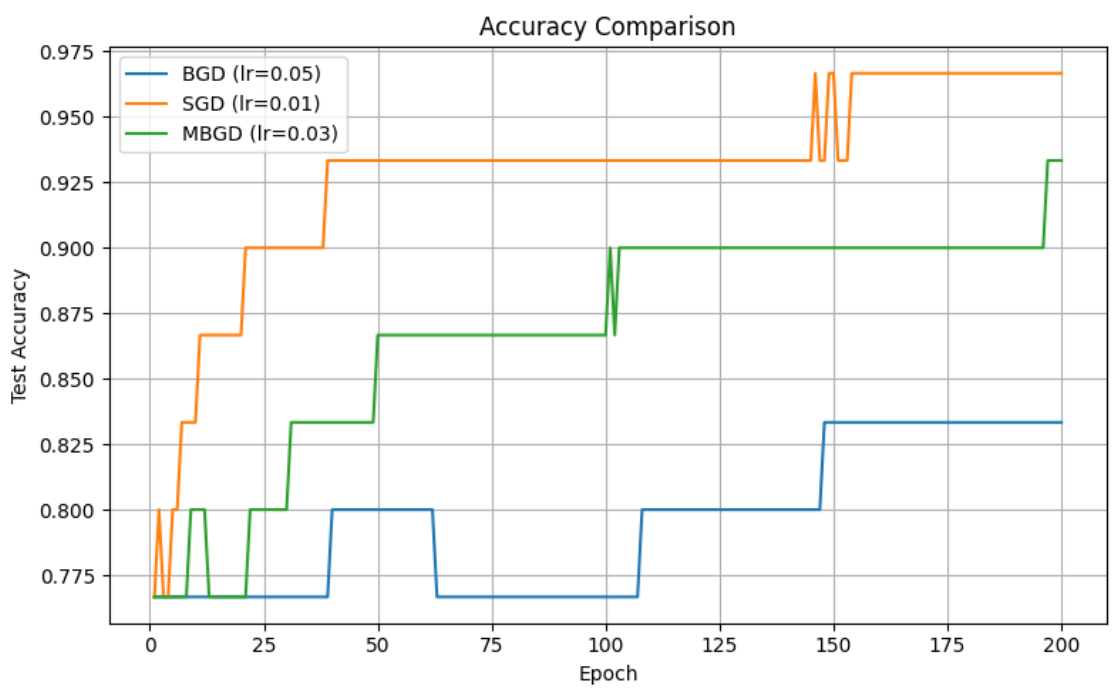
[9]:
```
  optimizer  learning_rate  final_train_loss  final_test_loss  \
0       bgd           0.05          0.320853         0.364136
1       sgd           0.01          0.079440         0.111994
2      mbgd           0.03          0.169326         0.214105

   final_train_acc  final_test_acc  avg_epoch_time  loss_std
0         0.925000        0.833333        0.000100  0.142892
1         0.975000        0.966667        0.001681  0.078096
2         0.958333        0.933333        0.000202  0.111898
```

1.10 9. Plot convergence and accuracy curves

[10]:
```
plot_loss_curves(histories)
```



[11]:
```
plot_accuracy_curves(histories)
```



1.11 10. Learning-rate sensitivity experiment

This section runs a small hyperparameter sweep to show how each optimizer reacts to different learning rates.

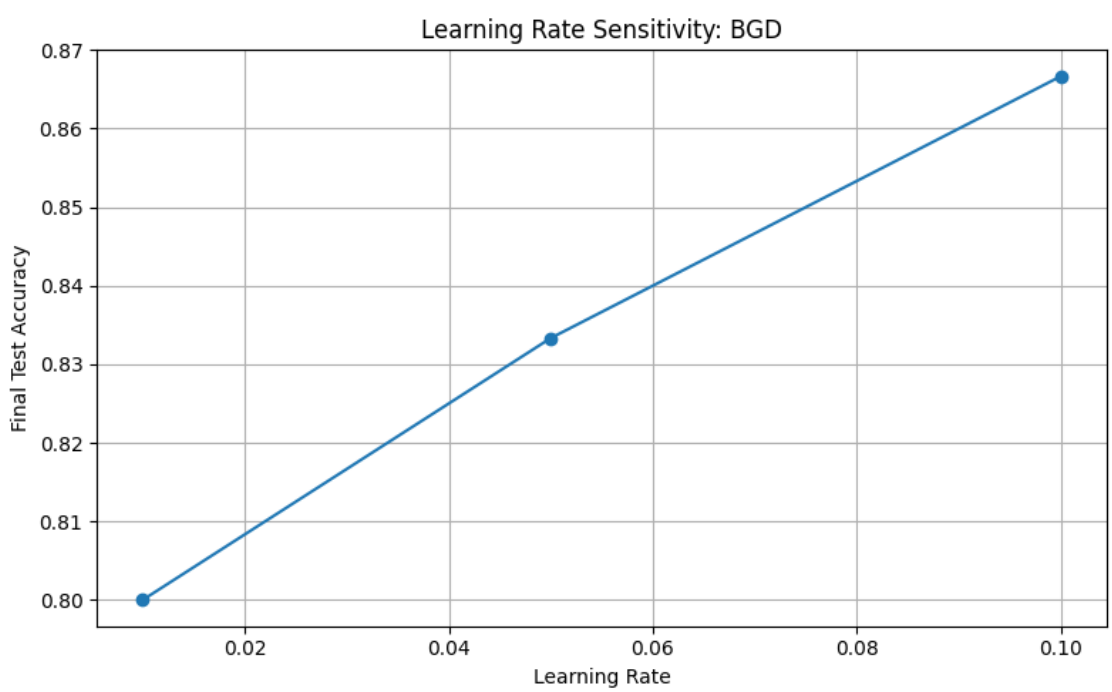
[12]:
```
lr_grid = {
    'bgd': [0.01, 0.05, 0.1],
    'sgd': [0.001, 0.01, 0.05],
    'mbgd': [0.01, 0.03, 0.05],
}

sweep_rows = []

for optimizer, lr_list in lr_grid.items():
    for lr in lr_list:
        _, history = train_model(
            X_train,
            y_train,
            X_test,
            y_test,
            optimizer=optimizer,
            learning_rate=lr,
            epochs=200,
            batch_size=16 if optimizer != 'sgd' else 1,
        )
        sweep_rows.append({
            'optimizer': optimizer,
            'learning_rate': lr,
            'final_train_loss': history['train_loss'][-1],
            'final_test_loss': history['test_loss'][-1],
            'final_train_acc': history['train_acc'][-1],
            'final_test_acc': history['test_acc'][-1],
            'avg_epoch_time': np.mean(history['epoch_time']),
        })

sweep_df = pd.DataFrame(sweep_rows)
sweep_df.sort_values(['optimizer', 'learning_rate']).reset_index(drop=True)
```

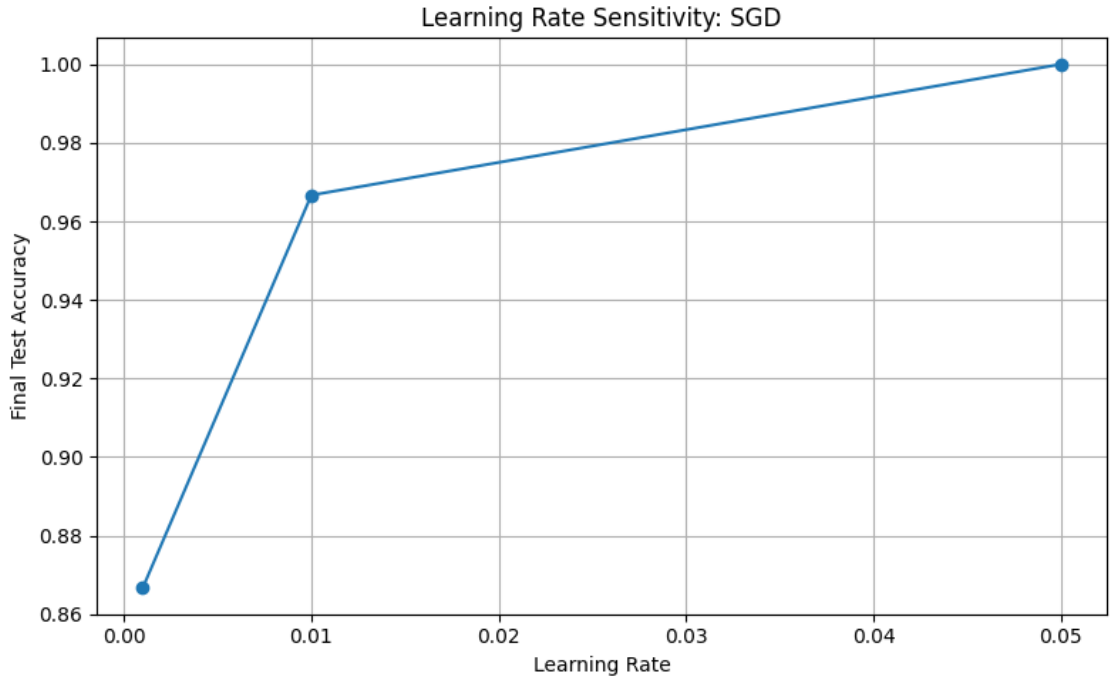
[12]:
```
  optimizer  learning_rate  final_train_loss  final_test_loss  \
0       bgd          0.010          0.518640         0.545970
1       bgd          0.050          0.320853         0.364136
2       bgd          0.100          0.251871         0.297899
3      mbgd          0.010          0.274479         0.319981
4      mbgd          0.030          0.169326         0.214105
5      mbgd          0.050          0.130785         0.172554
6       sgd          0.001          0.234178         0.280417
7       sgd          0.010          0.079440         0.111994
8       sgd          0.050          0.050110         0.066994
```

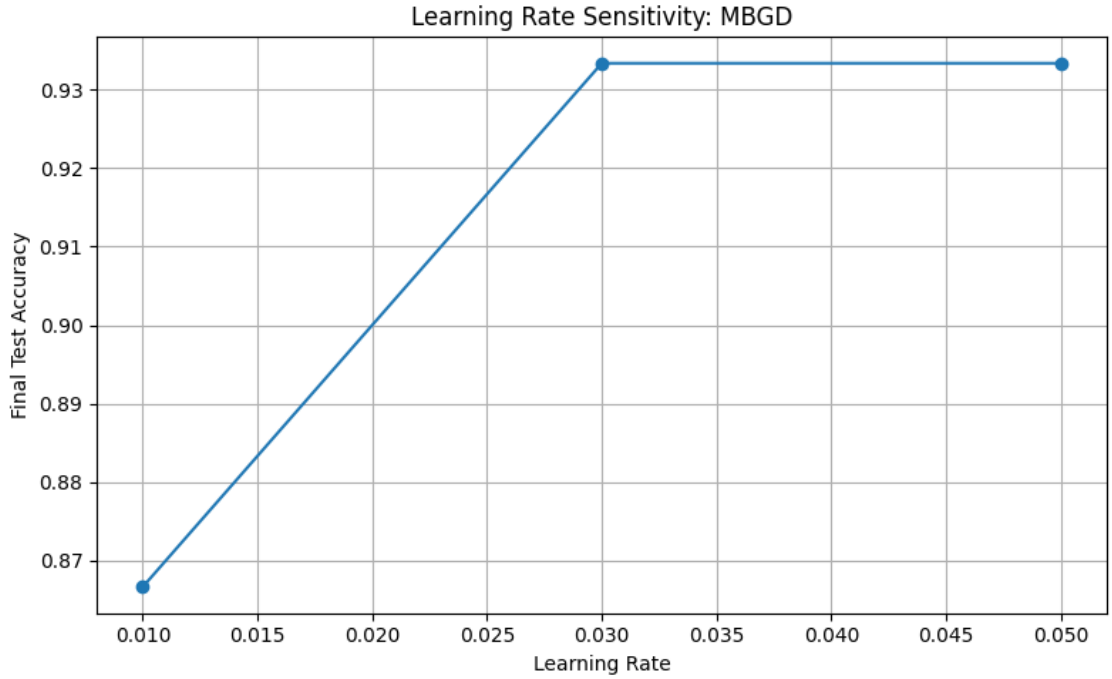
```
   final_train_acc  final_test_acc  avg_epoch_time
0         0.850000        0.800000        0.000111
1         0.925000        0.833333        0.000109
2         0.933333        0.866667        0.000101
3         0.925000        0.866667        0.000199
4         0.958333        0.933333        0.000198
5         0.966667        0.933333        0.000198
6         0.933333        0.866667        0.001652
7         0.975000        0.966667        0.001646
8         0.975000        1.000000        0.001642
```

1.12 11. Visualize test accuracy across learning rates

[13]:
```
for optimizer in ['bgd', 'sgd', 'mbgd']:
    subset = sweep_df[sweep_df['optimizer'] == optimizer].
 ↪sort_values('learning_rate')
    plt.figure()
    plt.plot(subset['learning_rate'], subset['final_test_acc'], marker='o')
    plt.xlabel('Learning Rate')
    plt.ylabel('Final Test Accuracy')
    plt.title(f'Learning Rate Sensitivity: {optimizer.upper()}')
    plt.grid(True)
    plt.tight_layout()
    plt.show()
```




Learning Rate Sensitivity: SGD
Final Test Accuracy
1.00
0.98
0.96
0.94
0.92
0.90
0.88
0.86
0.00
0.01
0.02
0.03
0.04
0.05
Learning Rate


Learning Rate Sensitivity: MBGD
Final Test Accuracy
0.93
0.92
0.91
0.90
0.89
0.88
0.87
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
0.050
Learning Rate

1.13 12. Short interpretation notes

You can adapt the following observations after running the notebook:

- All three optimization methods converge on the Iris dataset.
- SGD often reaches lower loss faster in epoch-based comparison because it updates parameters more frequently.
- BGD tends to be smoother but slower.
- MBGD usually provides a compromise between computational efficiency and predictive performance.
- Learning rate strongly affects both convergence speed and final accuracy.